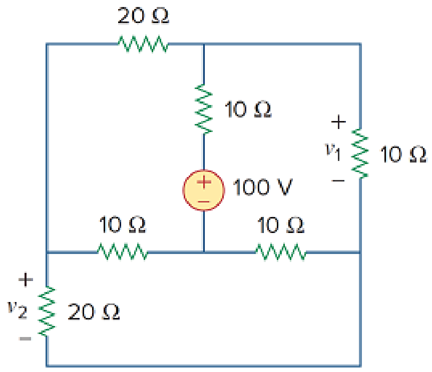
Chapter 3, Problem 56P

Problem

Determine $v1$ and $v2$ in the circuit of Fig. 3.101.

Figure 3.101:



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 3.101 in the text book.

Identify the loops and currents in the circuit diagram shown in Figure 1.

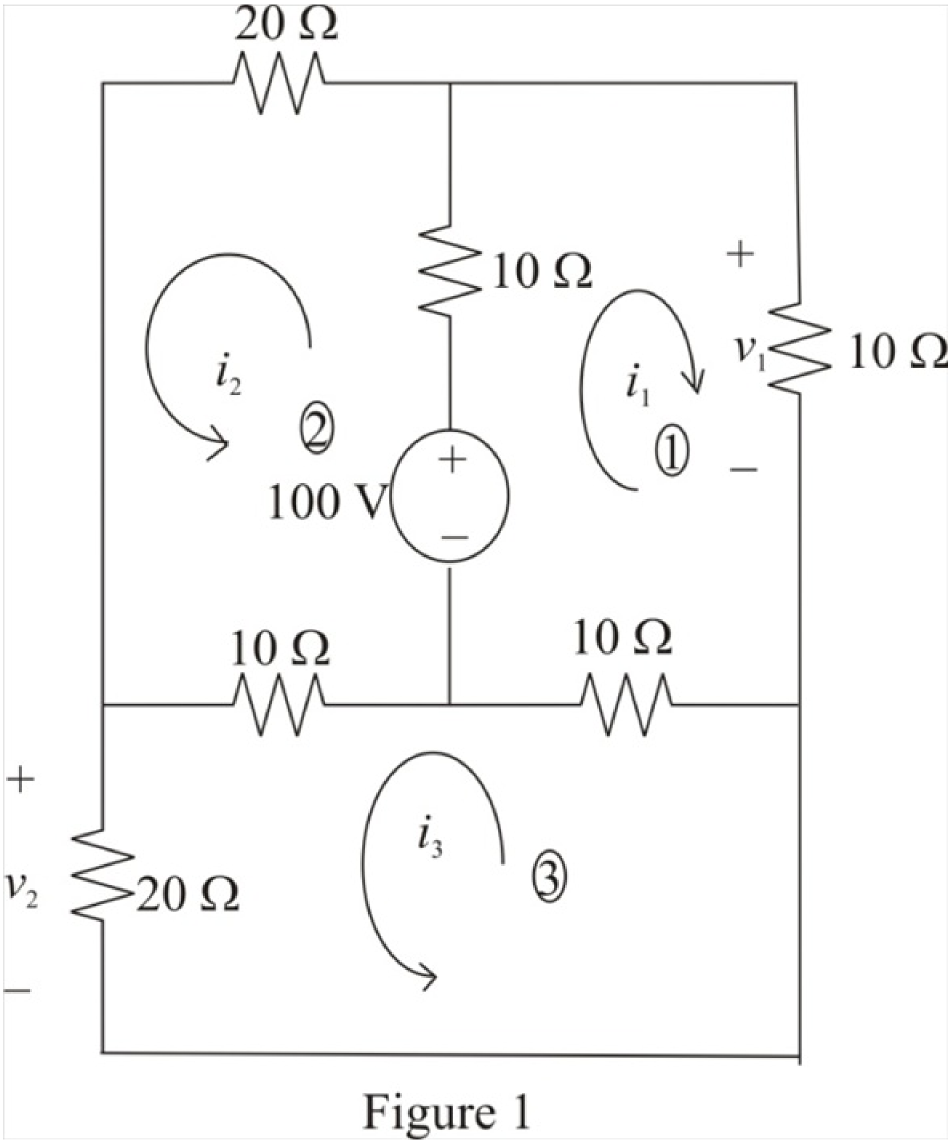


Figure 1

Step 2 of 7

Apply Kirchhoff's Voltage Law to Loop-1.

$$-100+(i_1+i_2)(10)+i_1(10)+(i_1+i_3)(10)=0$$
$$-100+(10+10+10)i_1+10i_2+10i_3=0$$
$$30i_1+10i_2+10i_3=100$$
$$3i_1+i_2+i_3=10$$
$$i_2=10-i_3-3i_1 \quad \ldots\ldots \text{(1)}$$

Step 3 of 7

Apply Kirchhoff's Voltage Law to Loop-2.

$$-100+(i_1+i_2)(10)+i_2(20)+(i_2-i_3)(10)=0$$
$$-100+10i_1+(10+20+10)i_2-10i_3=0$$
$$10i_1+40i_2-10i_3=100$$
$$i_1+4i_2-i_3=10$$

Substitute $10-i_3-3i_1$ for i_2.

$$i_1+4(10-i_3-3i_1)-i_3-10=0$$
$$i_1+40-4i_3-12i_1-i_3-10=0$$
$$-11i_1-5i_3+30=0$$
$$5i_3=30-11i_1$$
$$i_3=\frac{30-11i_1}{5} \quad \text{...... (2)}$$

Step 4 of 7

Apply Kirchhoff's Voltage Law to Loop-3.

$$(i_3-i_2)(10)+i_3(20)+(i_3+i_1)(10)=0$$
$$(10+20+10)i_3-10i_2+10i_1=0$$
$$40i_3-10i_2+10i_1=0$$
$$4i_3-i_2+i_1=0$$

Substitute $10-i_3-3i_1$ for i_2.

$$4i_3-(10-i_3-3i_1)+i_1=0$$
$$4i_3-10+i_3+3i_1+i_1=0$$
$$5i_3-10+4i_1=0$$

Substitute $\frac{30-11i_1}{5}$ for i_3.

$$5\left(\frac{30-11i_1}{5}\right)-10+4i_1=0$$
$$30-11i_1-10+4i_1=0$$
$$20-7i_1=0$$
$$7i_1=20$$
$$i_1=\frac{20}{7}$$
$$=2.857\text{ A}$$

Step 5 of 7

Calculate the value of the voltage v_1.

$$v_1=i_1(10)$$

Substitute 2.857 A for i_1.

$$v_1=(2.857\text{ A})(10\ \Omega)$$
$$=28.57\text{ V}$$

Therefore, the value of the voltage v_1 is, $\boxed{28.57\text{ V}}$.

Step 6 of 7

Recall equation (2).

$$i_3 = \frac{30 - 11i_1}{5}$$

Substitute $2.857\ \text{A}$ for i_1.

$$\begin{aligned} i_3 &= \frac{30 - (11)(2.857)}{5} \\ &= \frac{30 - 31.427}{5} \\ &= \frac{-1.427}{5} \\ &= -0.2857\ \text{A} \end{aligned}$$

Step 7 of 7

Calculate the value of the voltage v_2.

$$v_2 = i_3(20)$$

Substitute $-0.2857\ \text{A}$ for i_3.

$$\begin{aligned} v_2 &= (-0.2857\ \text{A})(20\ \Omega) \\ &= -5.714\ \text{V} \end{aligned}$$

Therefore, the value of the voltage v_2 is $\boxed{-5.714\ \text{V}}$.